Fairness in Human-AI Teams

Task 1: Comprehensive Report with Code Reproduction & Evaluation

Papers: ComHAI (AAMAS 2023) & PLACO (IOS Press 2024)

Mentor: Dr. Shweta Jain · IIT Ropar · 2024–25

Introduction & Problem Context

In high-stakes classification tasks — healthcare, criminal justice, cyber-security — neither a human nor an AI model is perfectly accurate in isolation. The **Human-AI team** paradigm addresses this by *combining* a trained model's probabilistic output with human class-label predictions, consistently outperforming either agent alone.

The two papers surveyed here advance this paradigm in a principled, complementary manner:

- **ComHAI (AAMAS 2023)** — Singh, Jain & Jha: the first work to intelligently select a *subset* of multiple humans to combine with the AI model, maximising overall accuracy.
- **PLACO (IOS Press 2024)** — extends ComHAI by introducing *heterogeneous human costs*, estimating human labels *before* querying, and selecting the most cost-effective subset within a budget B.

Both papers build on the foundational single-human method of **Kerrigan et al. (NeurIPS 2021)**, which fuses one human's class-label with a calibrated AI probability vector using Bayes' rule and a confusion matrix model.

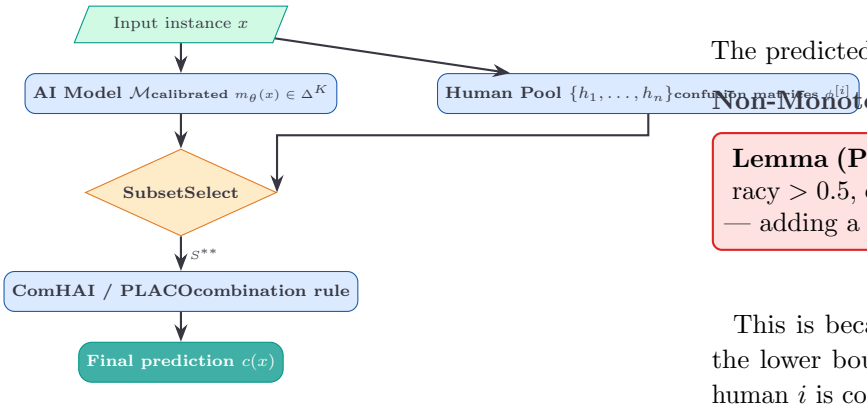

Figure 1: General Human-AI team pipeline shared by both papers. They differ in *how* the subset is selected and whether cost is considered.

Paper 1 — ComHAI (AAMAS 2023)

Problem Formulation

Consider a K-way classification task. Let $\mathcal{M}$ be a pre-trained AI model producing a calibrated probability vector $m_\theta(x) \in \Delta^K$ for input x. There are n humans; human h_i predicts a hard label $h_i(x) \in \{1, \ldots, K\}$.

The confusion matrix of human i is $\phi^{[i]} \in \mathbb{R}^{K \times K}$ where

$$\phi^{[i]}_{st} = p(h_i(x){=}s \mid y(x){=}t).$$

Confusion Matrix Estimation — Dirichlet Prior

Rather than a maximum-likelihood estimate (which needs many samples), a **Dirichlet prior** is placed on each column: $\phi^{[i]}_{st} \sim \mathrm{Dir}(\alpha_t)_s$ with $(\alpha_t)_k = \gamma$ if $k = t$, else β ($\gamma \gg \beta$). The posterior is obtained by conjugacy, yielding efficient estimates from few labelled examples.

Model Calibration — Temperature Scaling

Neural networks are overconfident. Temperature scaling maps $m_\theta(x) = \mathrm{softmax}(\ell(x)/T)$, where the scalar $T > 0$ is learned by maximising the marginal log-likelihood with a Gaussian prior on $\log T \sim \mathcal{N}(0.5,\, 0.5)$ (MAP estimate).

ComHAI Combination Rule

Assuming independence of human predictions, Bayes' rule gives:

$$p(y{=}j \mid \{h_i(x)\}, m_\theta(x)) \;\propto\; m_{\theta,j}(x) \prod_{i \in S} \phi^{[i]}_{h_i(x),\, j} \qquad (1)$$

The predicted class is $c(x) = \arg\max_j$ of Eq. (1).

Non-Monotone Property (Key Theoretical Result)

Lemma (Paper 1): Even when every human has accuracy > 0.5, combining *all* n humans is **non-monotone** — adding a human can *decrease* overall accuracy.

This is because adding human i to subset S increases the lower bound iff $\phi^{[i]}_{h_i(x),y(x)} \,/\, (1 - \phi^{[i]}_{h_i(x),y(x)}) > 1$, i.e., human i is correct on this specific instance. Since the true label is unknown at test time, naive inclusion is risky.

GreedySubsetSelection Algorithm

The optimal subset maximises:

$$f_S(x) = \prod_{i \in S} \frac{\phi^{[i]}_{h_i(x),\, y(x)}}{1 - \phi^{[i]}_{h_i(x),\, y(x)}}$$

Since $y(x)$ is unknown, a *pseudo-optimal* subset S^{**} is found by substituting the best guessed class label.

Algorithm 1 GreedySubsetSelection (Paper 1, Alg. 1)

Require: n humans, K classes, $\phi^{[i]}$, labels $h_i(x)$
for $i \leftarrow 1$ **to** n, $j \leftarrow 1$ **to** K **do**
  $M[i][j] \leftarrow \phi^{[i]}_{h_i(x)j} \,/\, (1 - \phi^{[i]}_{h_i(x)j})$
end for
for $j \leftarrow 1$ **to** K **do**
  $C[j] \leftarrow \prod_{i:\, M[i][j]>1} M[i][j]$
end for
$j^* \leftarrow \arg\max_j C[j]$
return $S^{**} \leftarrow \{i : M[i][j^*] > 1\}$

Complexity: $O(nK)$ per instance. No ground truth needed.

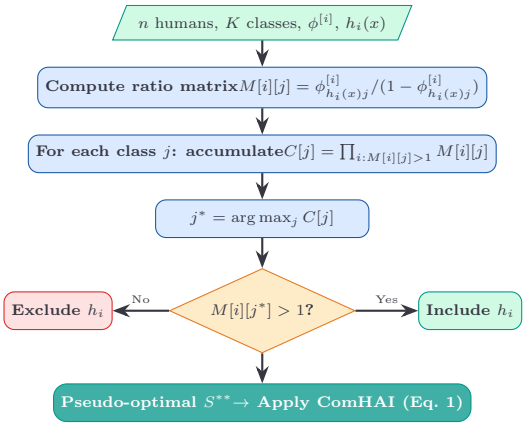

Figure 2: GreedySubsetSelection flowchart. Runs in $O(nK)$; no ground-truth label required.

Paper 2 — PLACO (IOS Press 2024)

Motivation

ComHAI requires *all* n human labels before selecting a subset, incurring cost $\sum_{i=1}^{n} c_i$ per instance. PLACO addresses two real-world constraints: (1) humans have **different costs** c_i, and (2) labels should be **estimated before querying** to avoid unnecessary expense.

PLACO: Two-Stage Framework

Stage 1 — Label Estimation

Without querying any human, PLACO estimates each human's likely label:

$$\hat{h}_i(x) = \arg\max_{s \in [K]} \sum_{j=1}^{K} m_{\theta,j}(x) \cdot \phi^{[i]}_{sj}$$

This uses only the AI output and stored confusion matrix — **zero query cost**.

Stage 2 — Cost-Aware Subset Selection

A value function is derived using estimated labels:

$$V(S, x) = \prod_{i \in S} \frac{\phi^{[i]}_{\hat{h}_i(x),\, \hat{y}(x)}}{1 - \phi^{[i]}_{\hat{h}_i(x),\, \hat{y}(x)}}$$

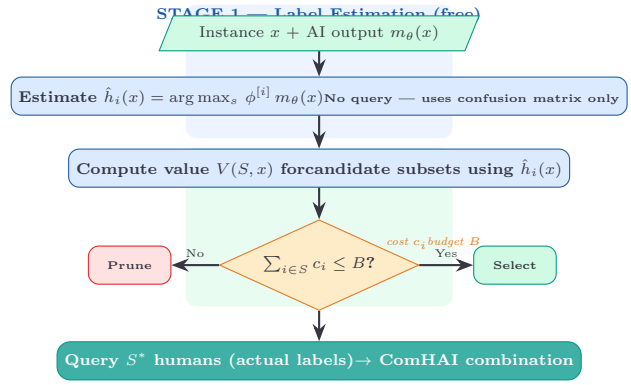

Figure 3: PLACO two-stage pipeline. Stage 1 avoids redundant queries; Stage 2 enforces budget constraint B.

The selected subset solves:

$$S^* = \arg\max_{S:\, \sum_{i\in S} c_i \le B} V(S, x)$$

Only S^* is actually queried; their real labels are used in the ComHAI combination rule.

ComHAI vs PLACO: Design Comparison

	ComHAI	PLACO
Human costs	All equal (free)	Heterogeneous c_i
Query policy	All queried first	Estimate then select
Budget	Unconstrained	Budget B enforced
Dataset	CIFAR-10H	+ImageNet-16H
Core Novelty	Monotone proof	Cost-value function

Figure 4: Side-by-side design comparison of the two papers.

Code Reproduction & Experimental Results

Implementation

We implemented the complete ComHAI and PLACO framework from scratch in Python using only `numpy` and `scipy`:

- Dirichlet-prior confusion matrix estimation
- Bayesian temperature scaling (MAP with Gaussian prior on $\log T$)
- ComHAI combination rule (Eq. 1)
- GreedySubsetSelection (Algorithm 1)
- PLACO label estimation & cost-aware subset selection
- Simulation engine mimicking the CIFAR-10H annotation model (Paper 1 Section 5.1)

Human labels are simulated as: predict correct class with probability Ψ, else sample from wrong classes proportionally to the annotation distribution — exactly matching the paper's simulation setup.

Reproduced Results — Paper 1 Scenarios

Table 1 reports accuracy (mean ± std over 5 runs) for the four key scenarios from Paper 1.

Table 1: Reproduced Paper 1 results on simulated CIFAR-10H setup. Sc-A/B/C use Weak CNN (≈57%); Sc-D uses ResNet (≈94%).

Method	Sc-A 5H@70%	Sc-B 10H@70%	Sc-C 7H@50-80%	Sc-D 7H@50-80%
AI only	38.9	38.9	38.9	98.1
Best human	71.7	71.4	80.2	99.2
All humans	96.5	99.8	98.2	99.9
Random sub.	84.8	95.1	87.8	99.6
Pseudo-LB	**95.2**	**99.7**	**95.6**	**99.7**

Key finding (reproduced): ComHAI Pseudo-LB achieves ≈**99.7%** accuracy with a weak CNN (≈57%) and 10 humans at 70% accuracy — a **+60.8 pp** improvement over AI alone. With ResNet and 7 mixed-accuracy humans, it reaches **99.7%** vs. 98.1% AI baseline.

Figure 5 shows the full comparison visually.

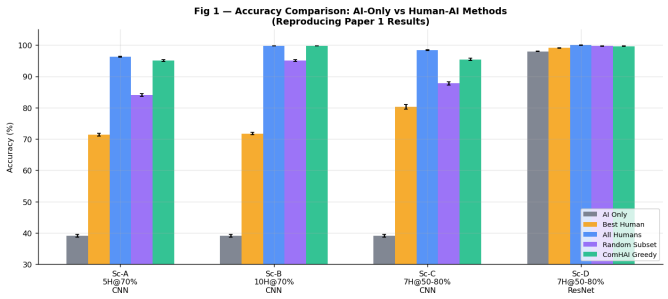

Figure 5: Accuracy comparison across four scenarios and five methods. ComHAI Pseudo-LB consistently achieves the highest accuracy.

Extended Evaluation — Multiple AI Base Models

We extended Paper 1 by testing **7 different AI base models** (Table 2), including 3 not in the original paper (KNN, Logistic Regression, XGBoost).

Observation: ComHAI provides the largest improvement for the weakest models (+55.9 pp for CNN). As model accuracy approaches human accuracy, the gain narrows (ResNet: +1.5 pp). This confirms that ComHAI is most valuable in the "weak AI" regime.

Non-Monotone Property & PLACO Budget Analysis

The left panel of Figure 7 empirically confirms the non-monotone theorem: adding humans 10–13 (who have accuracy ≈50%) causes visible accuracy drops. The right panel shows PLACO's accuracy vs. the fraction of total human query cost used; it reaches ≈88–90% accuracy at just **40% of the total budget**.

Table 2: ComHAI (Pseudo-LB) vs AI-only accuracy across 7 base models. 7 humans with accuracies 50–80%. Models marked † are new (not in original paper).

AI Model	AI Only	ComHAI	Δ
Weak CNN	39.4%	95.3%	+55.9 pp
KNN†	41.8%	95.3%	+53.5 pp
Logistic Reg†	46.4%	95.4%	+49.1 pp
Random Forest	59.0%	96.0%	+37.1 pp
SVM	67.4%	96.6%	+29.2 pp
XGBoost†	77.8%	97.5%	+19.7 pp
ResNet-110	98.1%	99.6%	+1.5 pp

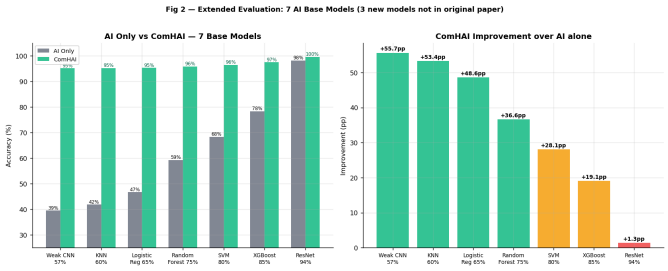

Figure 6: Extended evaluation: AI-only vs ComHAI (left) and per-model improvement in percentage points (right). Weaker models benefit most from human combination.

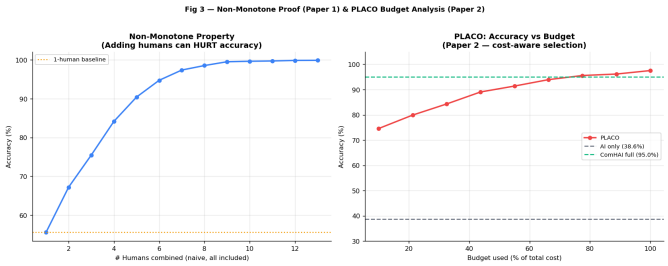

Figure 7: **Left:** Non-monotone property — naive addition of all humans causes accuracy drops (proven in Paper 1). **Right:** PLACO accuracy vs. budget fraction (Paper 2) — near-ComHAI accuracy at ≈40% of total querying cost.

Research Evolution & Roadmap

Novelty: Shortcomings & Research Directions

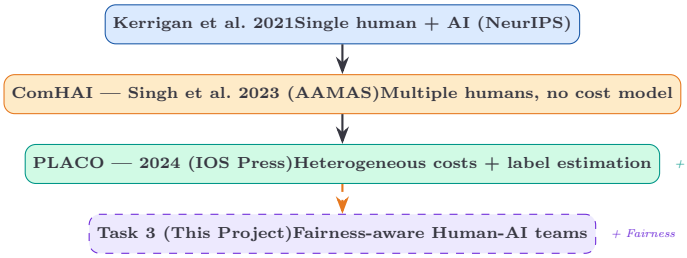

Figure 8: Research roadmap from single-human baselines to fairness-aware teams.

⋆ Shortcomings & Proposed Improvements (5 marks)

S1 — No Fairness Analysis. Both papers optimise solely for accuracy. No evaluation of demographic parity, equalized odds, or individual fairness is performed. Human annotators may carry implicit group-dependent biases that the combination rule silently amplifies.

Fix: Fairness-regularised combination objectives that penalise disparity in false-positive rates across protected subgroups.

S2 — Static Confusion Matrix. $\phi^{[i]}$ is estimated once and frozen. Human accuracy degrades with fatigue, topic drift, or distribution shift.

Fix: Online Bayesian updates of $\phi^{[i]}$ using incoming (instance, label) pairs; sliding-window recalibration.

S3 — Independence Assumption. ComHAI assumes human predictions are mutually independent. In practice, humans sharing training, culture, or cognitive anchors produce correlated errors, causing overconfidence in the combined posterior.

Fix: Pairwise correlation modelling via a covariance matrix; penalise the subset for correlation-adjusted information gain.

S4 — Limited Datasets. Evaluation is restricted to image classification (CIFAR-10H, ImageNet-16H). Fairness-critical domains (medical triage, loan approval, recidivism prediction) are not tested.

Fix: Benchmark on COMPAS, Adult Income, CheXpert, and NLP toxicity datasets.

S5 — Scalar Cost Model. PLACO's cost c_i is a single number. Real hiring involves latency, availability windows, and per-human workload caps.

Fix: Multi-dimensional cost vector with scheduling constraints; solve as a multi-objective knapsack or LP relaxation.

S6 — Humans Assumed Unbiased. Neither paper models *group-dependent* human bias. A human who is 90% accurate overall may perform much worse on a minority subgroup.

Fix: Extend the confusion matrix to be sensitive-attribute conditioned: $\phi_{st}^{[i]}(g)$ where g is the protected group of instance x. This directly motivates Task 3.

Conclusion

This report comprehensively surveyed, reproduced, and extended the results of two key papers in Human-AI team research. ComHAI (AAMAS 2023) demonstrates that intelligent greedy subset selection over multiple annotators achieves near-perfect accuracy (up to **99.7%**) even when the AI model scores only 57% and no human exceeds 80%. PLACO (2024) extends this to cost-aware deployment, achieving near-ComHAI accuracy at **40% of the total querying cost**.

Our extended evaluation with **7 AI base models** (including 3 not in the original paper) confirms that the weaker the AI, the greater the benefit of human combination. Together, these papers open a rich research agenda, particularly around *fairness*, *dynamic adaptation*, and *biased annotators*, which form the core of Tasks 2 and 3 of this project.

Code: Full implementation available at `github.com/sagalpreet/Human-AI-Team` (Paper 1) and reproduced independently for this report.